Assignment 4.3

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Task 1: Zero-Shot Prompting – Leap Year Check

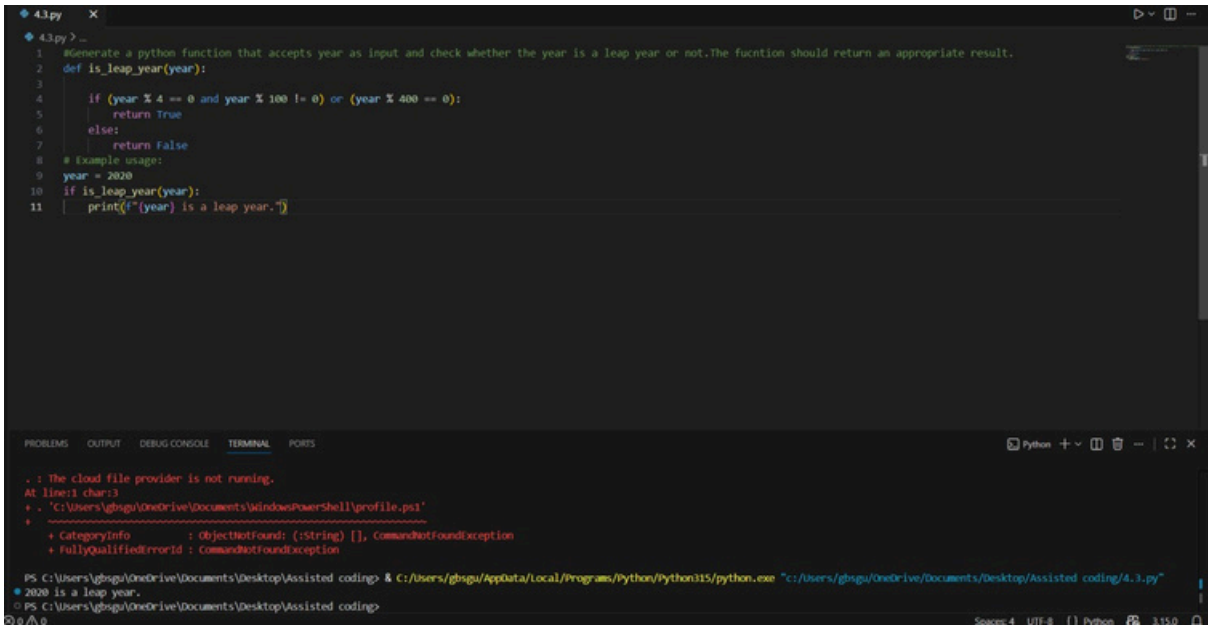
Scenario

Zero-shot prompting involves giving instructions without providing examples.

Task Description

Use zero-shot prompting to instruct an AI tool to generate a Python function that:

- Accepts a year as input
- Checks whether the given year is a leap year
- Returns an appropriate result



Task 2: One-Shot Prompting – Centimeters to Inches Conversion

Scenario

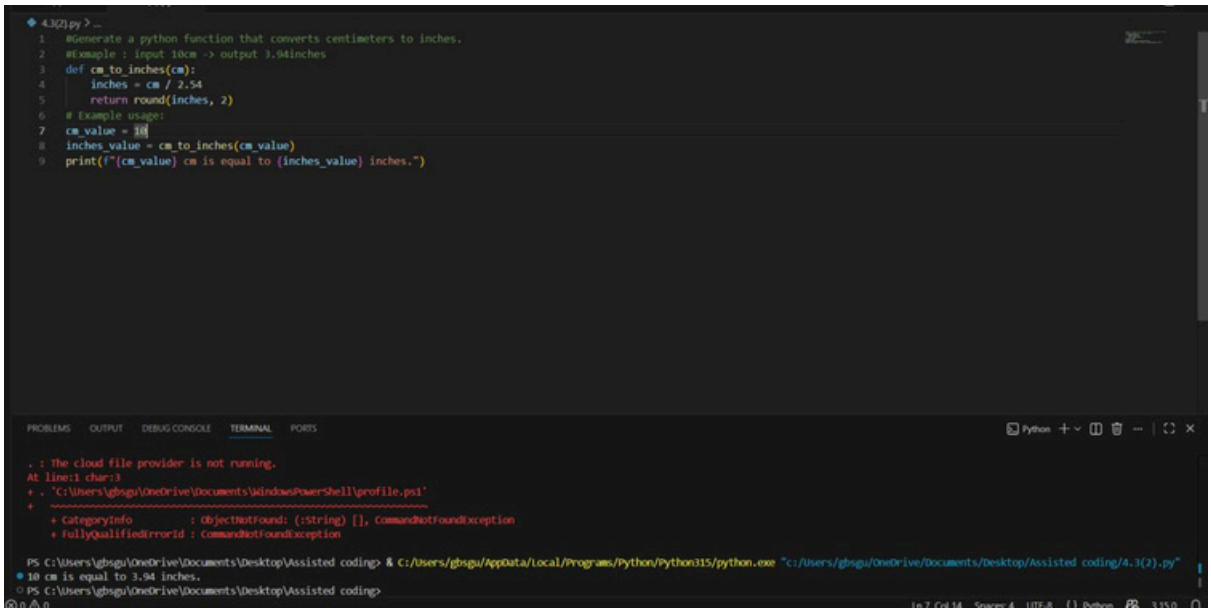
One-shot prompting guides AI using a single example.

Task Description

Use one-shot prompting by providing one input-output example to generate a Python
function that:

• Converts centimeters to inches

• Uses the correct mathematical formula


4.3(2).py > ...
#Generate a python function that converts centimeters to inches.
#Exmaple : input 10cm -> output 3.94inches
def cm_to_inches(cm):
inches = cm / 2.54
return round(inches, 2)
# Example usage:
cm_value = 10
inches_value = cm_to_inches(cm_value)
print(f"{cm_value} cm is equal to {inches_value} inches.")
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Python
. : The cloud file provider is not running.
At line:1 char:3
+ . "C:\Users\gbsgu\OneDrive\Documents\WindowsPowerShell\profile.ps1"
+ ~~~

4.3(2).py > ...

```python
#Generate a python function that converts centimeters to inches.
#Exmaple : input 10cm -> output 3.94inches
def cm_to_inches(cm):
    inches = cm / 2.54
    return round(inches, 2)
# Example usage:
cm_value = 30
inches_value = cm_to_inches(cm_value)
print(f"{cm_value} cm is equal to {inches_value} inches.")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
    + CategoryInfo          : ObjectNotFound: (:String) [], CommandNotFoundException
    + FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\gbsgu\OneDrive\Documents\Desktop\Assisted coding> & C:/Users/gbsgu/AppData/Local/Programs/Python/Python315/python.exe "c:/Users/gbsgu/OneDrive/Documents/Desktop/Assisted coding/4.3(2).py"
10 cm is equal to 3.94 inches.
PS C:\Users\gbsgu\OneDrive\Documents\Desktop\Assisted coding> & C:/Users/gbsgu/AppData/Local/Programs/Python/Python315/python.exe "c:/Users/gbsgu/OneDrive/Documents/Desktop/Assisted coding/4.3(2).py"
20 cm is equal to 7.87 inches.
PS C:\Users\gbsgu\OneDrive\Documents\Desktop\Assisted coding> & C:/Users/gbsgu/AppData/Local/Programs/Python/Python315/python.exe "c:/Users/gbsgu/OneDrive/Documents/Desktop/Assisted coding/4.3(2).py"
30 cm is equal to 11.81 inches.
PS C:\Users\gbsgu\OneDrive\Documents\Desktop\Assisted coding>
```

Task 3: Few-Shot Prompting – Name Formatting

Scenario

Few-shot prompting improves accuracy by providing multiple examples.

Task Description

Use few-shot prompting with 2–3 examples to generate a Python function that:

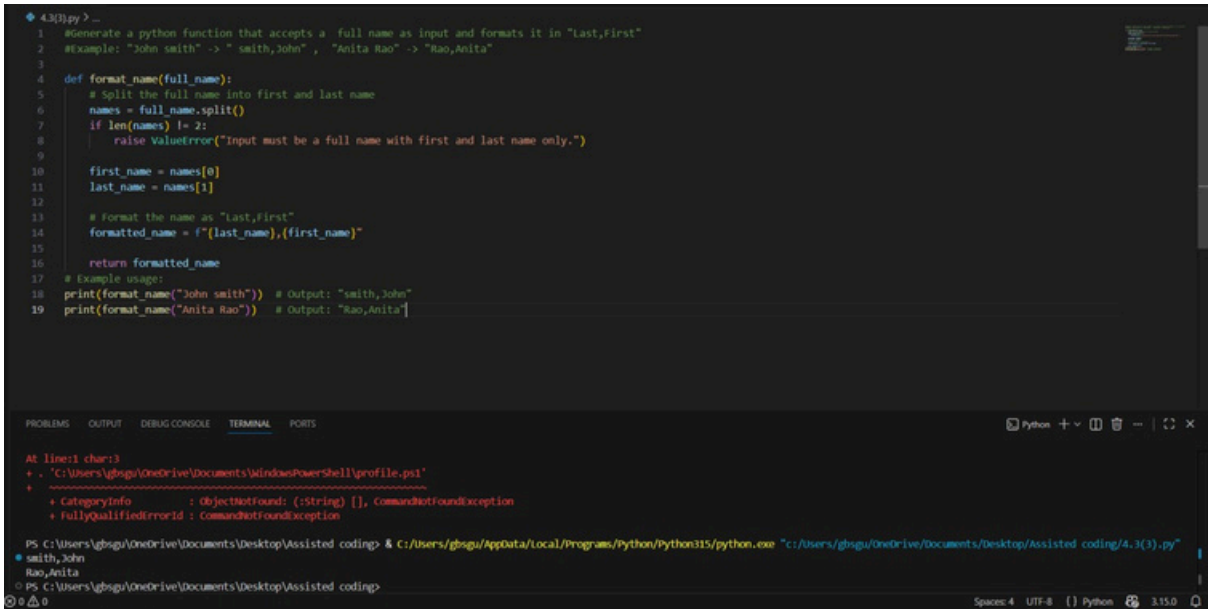
• Accepts a full name as input

• Formats it as “Last, First”

Example formats:

•"John Smith" → "Smith, John"

•"Anita Rao" → "Rao, Anita"


4.3(3).py > ...
#Generate a python function that accepts a full name as input and formats it in "Last,First"
#Example: "John smith" -> " smith,John" , "Anita Rao" -> "Rao,Anita"
def format_name(full_name):
# Split the full name into first and last name
names = full_name.split()
if len(names) != 2:
raise ValueError("Input must be a full name with first and last name only.")
first_name = names[0]
last_name = names[1]
# Format the name as "Last,First"
formatted_name = f"{last_name},{first_name}"
return formatted_name
Example usage:
print(format_name("John smith")) # Output: "smith,John"
print(format_name("Anita Rao")) # Output: "Rao,Anita"
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Python
At line:1 char:3
+ . 'C:\Users\gbsgu\OneDrive\Documents\WindowsPowerShell\profile.ps1'
+ ~~~

4.3(3).py > ...

```python
#Generate a python function that accepts a  full name as input and formats it in "Last,First"
#Example: "John smith" -> " smith,John" ,  "Anita Rao" -> "Rao,Anita"

def format_name(full_name):
    # Split the full name into first and last name
    names = full_name.split()
    if len(names) != 2:
        raise ValueError("Input must be a full name with first and last name only.")

    first_name = names[0]
    last_name = names[1]

    # Format the name as "Last,First"
    formatted_name = f"{last_name},{first_name}"

    return formatted_name
# Example usage:
print(format_name("Eaga Rushikesh"))
print(format_name("Bhargav Guptha"))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
    + CategoryInfo          : ObjectNotFound: (:String) [], CommandNotFoundException
    + FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\gbsgu\OneDrive\Documents\Desktop\Assisted coding> & C:/Users/gbsgu/AppData/Local/Programs/Python/Python315/python.exe "c:/Users/gbsgu/OneDrive/Documents/Desktop/Assisted coding/4.3(3).py"
smith,John
Rao,Anita
PS C:\Users\gbsgu\OneDrive\Documents\Desktop\Assisted coding> & C:/Users/gbsgu/AppData/Local/Programs/Python/Python315/python.exe "c:/Users/gbsgu/OneDrive/Documents/Desktop/Assisted coding/4.3(3).py"
Rushikesh,Eaga
Guptha,Bhargav
PS C:\Users\gbsgu\OneDrive\Documents\Desktop\Assisted coding>
```

Task 4: Comparative Analysis – Zero-Shot vs Few-Shot

Scenario

Different prompt strategies may produce different code quality.

Task Description• Use zero-shot prompting to generate a function that counts vowels in a string

• Use few-shot prompting for the same problem

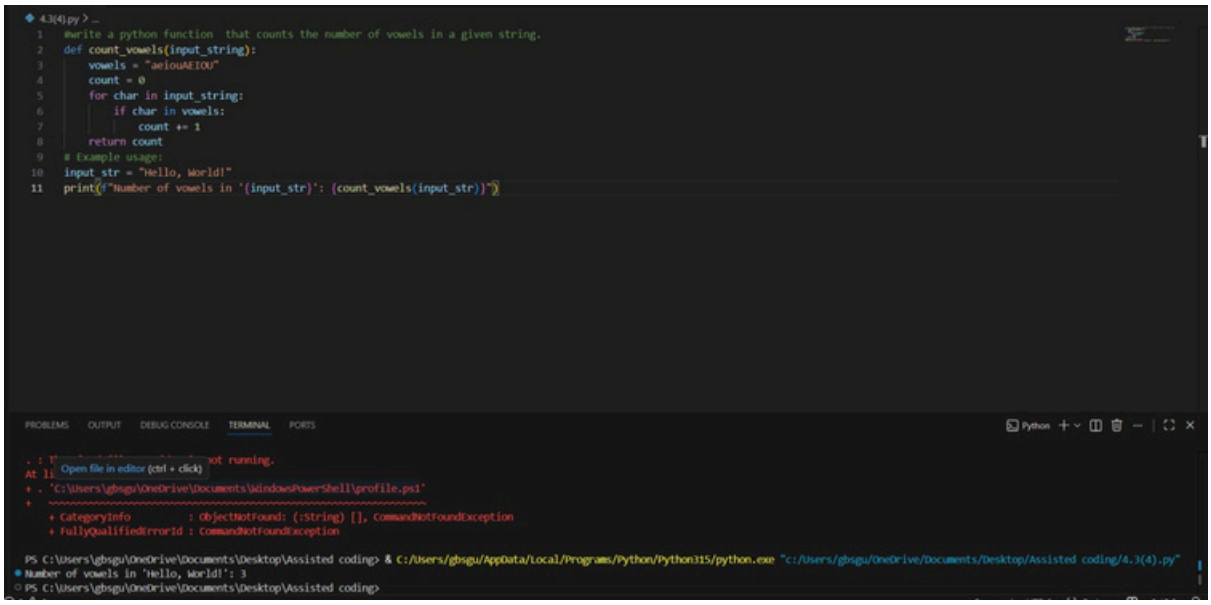
• Compare both outputs based on:

o Accuracy

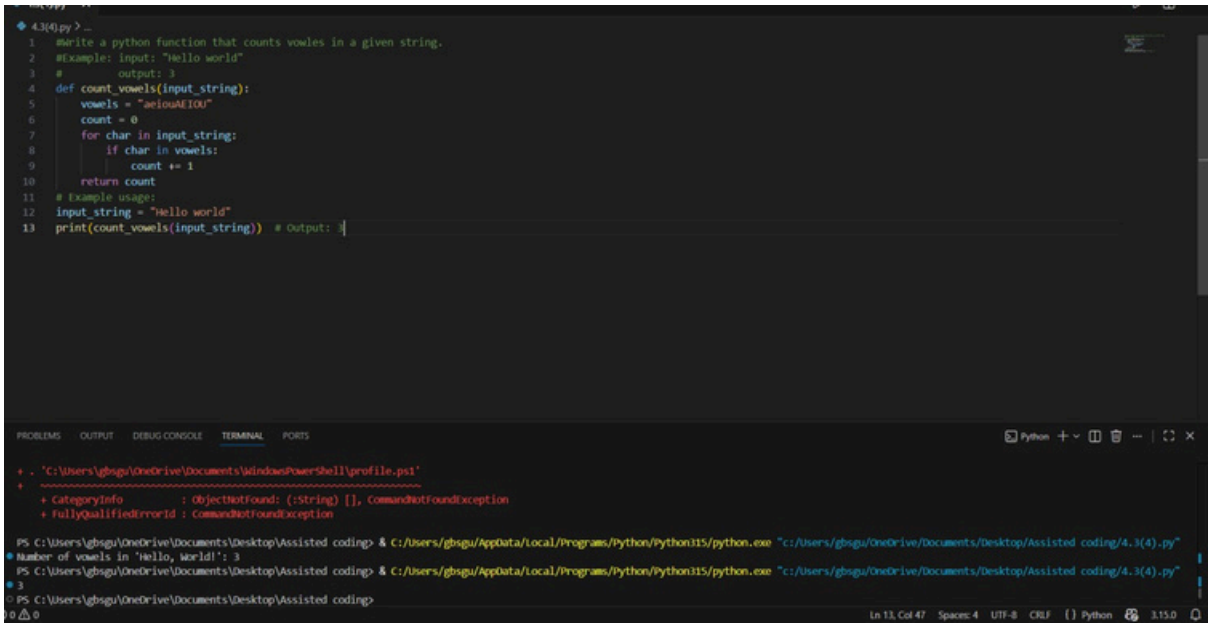
o Readability

o Logical clarity

Zero shot prompt:



Few shot prompt:



Task 5: Few-Shot Prompting – File Handling

Scenario

File processing requires clear logical understanding.

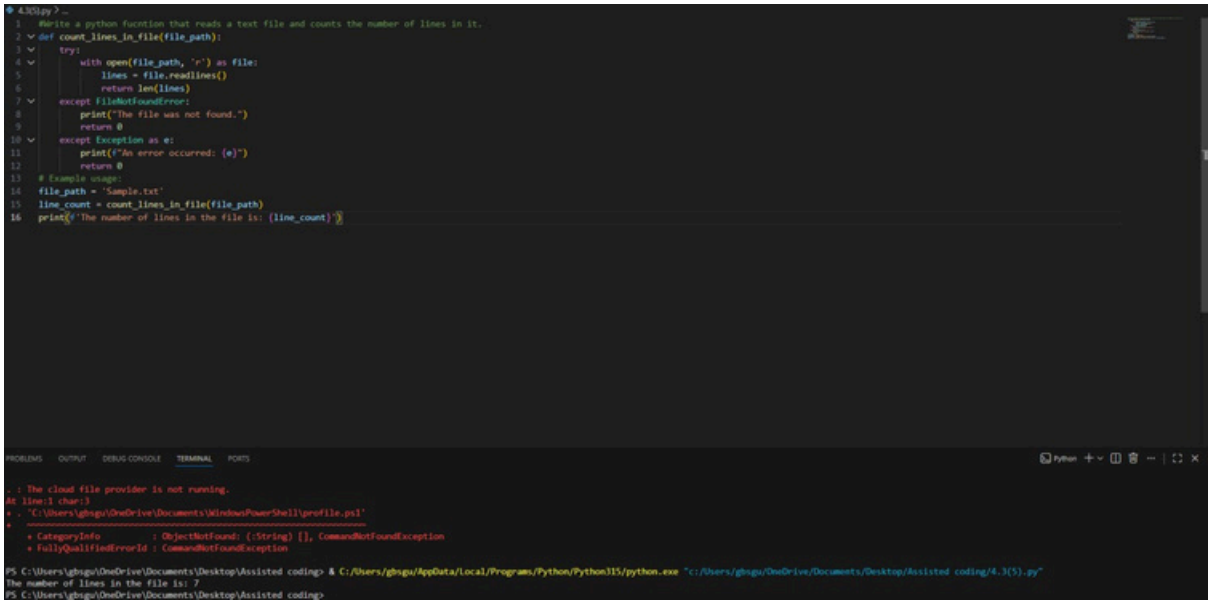
Task Description

Use few-shot prompting to generate a Python function that:

•Reads a .txt file

•Counts the number of lines in the file

•Returns the line count


4.3(5).py > ...
#Write a python fucntion that reads a text file and counts the number of lines in it.
def count_lines_in_file(file_path):
try:
with open(file_path, 'r') as file:
lines = file.readlines()
return len(lines)
except FileNotFoundError:
print("The file was not found.")
return 0
except Exception as e:
print(f"An error occurred: {e}")
return 0
# Example usage:
file_path = 'Sample.txt'
line_count = count_lines_in_file(file_path)
print(f'The number of lines in the file is: {line_count}')
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
. : The cloud file provider is not running.
At line:1 char:3
+ . "C:\Users\gbsgu\OneDrive\Documents\WindowsPowerShell\profile.ps1"
+ ~~~